Astro HW 12

Pierson Lipschultz

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12.1 Eddington luminosity and massive stars

a)

Given ...

$$L = 1.2L_\odot \left(\frac{M}{M_\odot}\right)^{3.92}$$

$$L_{edd} = \frac{4\pi G M_p c}{\sigma_e}$$

$$\sigma_e = 6.55 \times 10^{-29} m^2, \quad m_p = 1.67 \times 10^{-27} kg$$

$$\boxed{M_{max} = 3.514 \times 10^1 \, \mathrm{L}_\odot}$$

b)

This calculated value of $M_{max} = 3.514 \times 10^1 \, \mathrm{L}_\odot$ is much lower than a considerable amount of observed systems. For example, the current most massive star has a proposed mass of $346 \pm 42 M_\odot$[1]. All the most massive stars are Wolf-Rayet stars, meaning that they are so massive that they start to sherd their exterior hydrogen envelopes.

12.2 Accretion disk temperatures

a)

Given ...

$$T \approx \left[\frac{GM}{2\sigma_{SB} r^2} \Sigma v\right]^{1/4}$$

$$\dot{M} = 2\pi r \Sigma(r) v(r)$$

We find ...

$$\dot{M} = 2\pi r \Sigma(r) v(r)$$

$$\Sigma(r) v = \frac{\dot{M}}{r 2\pi}$$

$$T \approx \left[\frac{GM}{2\sigma_{SB} r^2} \frac{\dot{M}}{r 2\pi}\right]^{1/4}$$

$$T \approx \left[\frac{GM\dot{M}}{4\pi \sigma_{SB} r^2}\right]^{1/4}$$

b)

Given ...

$$L_e = 4\pi r^2 \sigma_{sb} T^4$$
$$r = \frac{6GM}{c^2}$$
$$r \propto M, \quad M \propto L_E$$

We can find ...

$$L_e = 4\pi \left(\frac{6GM}{c^2}\right)^2 \sigma_{sb} T^4$$

By proportionality ...

$$\Rightarrow M^2 T^4 \propto M$$
$$\Rightarrow T^4 \propto \frac{M}{M^2}$$
$$\boxed{T = M^{-1/4}}$$

c)

Given ...

$$\dot{M} = \dot{m}\left(\frac{4\pi G M m_p}{\sigma_e \eta c}\right)$$
$$\dot{m} = 1 \quad \eta \sim .1$$
$$r = \frac{(6GM)}{c^2}$$
$$T(r) \approx \left[\frac{GM\dot{M}}{4\pi\sigma_{SB} r^3}\right]^{1/4}$$

We find ...

$$\boxed{T_{4M_\odot} = 1.755 \times 10^7\,\text{K}, \quad T_{10^9 M_\odot} = 1.396 \times 10^5\,\text{K}}$$

d)

We find that they lie in the microwave and radio wave spectrum.

12.3 Stellar close encounters

a)

b)

$$t = \frac{1}{n_*(\pi R_{eff}^2)}$$

c)

Divide answer of above by 5

12.4 Expansion of a Newtonian universe

a)

Given ...

$$\frac{\dot{a}^2}{a^2} = \frac{8\pi G\rho^{c,0}}{3}\frac{1}{a^3}$$

We can calculate ...

$$\dot{a}^2 = \frac{8\pi G\rho^{c,0}}{3}\frac{1}{a}$$

$$\frac{8\pi G\rho^{c,0}}{3} = k$$

$$\dot{a} = \sqrt{\frac{k}{a}}$$

$$\frac{da}{dt} = \sqrt{\frac{k}{a}}$$

$$da = \sqrt{\frac{k}{a}}dt$$

$$\sqrt{a}da = \sqrt{k}dt$$

$$\int \sqrt{a}da = \int \sqrt{k}dt$$

$$\frac{2}{3}a^{3/2} = t\sqrt{k}$$

$$a = \left(\frac{3}{2}t\sqrt{k}\right)^{2/3}$$

$$\boxed{a(t) = \left(\frac{3}{2}t\sqrt{\frac{8\pi G\rho_{c,0}}{3}}\right)^{2/3}}$$

b)

c)

Given ...

$$\frac{1}{t_0} = \frac{1}{H_0}$$

We find ...

$$\boxed{T_{uni} = 13.97\,\mathrm{Gyr}}$$

12.5 Olbers's paradox again

a)

Given ...

$$n_{wd} = \frac{3H_0^2}{8\pi G}$$

We find

$$\boxed{n_{wd} = 6.613 \times 10^{-57}\,\mathrm{m}^3}$$

b)

Sources

[1] Z. Keszthelyi et al. "Evolutionary models for the very massive stars in the R136 cluster of 30 Doradus in the Large Magellanic Cloud". In: *Astronomy and Astrophysics* 700 (Aug. 2025), A186. ISSN: 1432-0746. DOI: 10.1051/0004-6361/202554758. URL: http://dx.doi.org/10.1051/0004-6361/202554758.